Highlights

Sparse defect localization - A Periodic Empirical Operator Framework for Inverse Scattering and Structural Health Monitoring

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- Presents a fully vectorized linear operator framework for inverse scattering in NDT.
- Relies solely on the principles of linear time-invariant (LTI) systems, avoiding the need for theoretical assumptions about wave propagation.
- Enforces periodic discrete time, enabling FFT-diagonalizable circular convolution operators which are solvable using the Tikhonov regularized inverse.
- Localizes defects in both simulated aluminum plates and real-world CFRP plates.
- Post-calibration, only sparse sensor measurements are needed for rapid health monitoring.

Sparse defect localization - A Periodic Empirical Operator Framework for Inverse Scattering and Structural Health Monitoring

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ABSTRACT

We present an empirically calibrated, fully vectorized linear-operator framework for inverse scattering, motivated by challenges in non-destructive testing (NDT) and structural health monitoring (SHM) where reliable forward models are difficult to obtain and maintain. The method enables defect localization without requiring an explicit physical model of wave propagation. Source–receiver impulse-response operators are constructed through a one-time calibration using bonded piezoelectric transducers for excitation and dense Laser Doppler Vibrometer (LDV) measurements. Enforcing discrete periodic time behavior represents propagation as circular convolution, yielding circulant operators diagonalizable via the discrete Fourier transform (DFT) and allowing defect localization to be posed as a Tikhonov-regularized linear inverse problem using only sparse monitoring data.

Experimental validation on a bidirectional carbon-fiber-reinforced polymer (CFRP) plate demonstrates localization of a 1 cm³ adhered defect using measurements acquired solely at transducer locations after calibration. Computational cost is concentrated in the calibration stage, while subsequent evaluations reduce to rapid matrix–vector operations. Additional simulations complement the experimental results and probe performance across defect scenarios and sensing configurations, including proof-of-concept localization in a single-transducer (pulse-echo) setting.

Subject to the stated linear time-invariant assumptions and modeling choices, the framework provides a data-driven alternative for NDT and SHM when high-fidelity digital twins are difficult to maintain, and it may be adaptable to other wave-based sensing applications.

1. Introduction

Structural Health Monitoring (SHM) of engineered components seeks to localize and characterize defects before they compromise structural integrity. The importance of SHM is closely tied to the cost of failure, which is particularly high in aviation, and has motivated extensive research on monitoring approaches for relatively simple, flat structures [34]. Among available techniques, piezoelectric transducers are commercially attractive because of their low cost and ease of integration. Nevertheless, significant difficulties arise when these concepts are transferred to carbon-fiber-reinforced polymers (CFRPs) [19], and a range of dedicated strategies for CFRP inspection has therefore been investigated [14, 31]. Interpreting guided-wave measurements in such materials is inherently complex: overlapping modes often require dedicated processing, such as ridge-based time–frequency analysis, beamforming, or migration, to extract meaningful arrivals [27], while benign geometric features, multiple reflections, and mode conversions at joints and thickness transitions generate coherent clutter that can obscure damage signatures [4, 24]. These effects are amplified by the pronounced anisotropy, laminate heterogeneity, and viscoelastic damping of CFRPs, which lead to direction- and frequency-dependent dispersion and attenuation [3, 25]. Curvature, stiffeners, and adhesive interfaces introduce additional localized scattering, and in large curved assets such as rocket fuselages or wind-turbine blades, spatially varying thickness and fiber orientation challenge travel-time-based concepts [35, 7, 18, 23, 26].

Sparse monitoring of Lamb waves with a limited number of piezoelectric sensors [32] would therefore be highly attractive. However, such approaches generally rely on detailed numerical forward models. Modern numerical frameworks, including finite- and spectral-element solvers combined with higher-order plate formulations, have proven capable of reproducing dispersion, mode conversion, and multi-path scattering in anisotropic laminates with high fidelity while remaining computationally efficient at ultrasonic frequencies [3, 20]. Nevertheless, such models are

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difficult to maintain for composite structures [30, 21]. CFRP components, in particular, exhibit variability arising from lay-up and manufacturing processes that undermines the reliability of predictive digital twins [6, 11]. In response to these limitations, data-driven strategies based on Laser Doppler Vibrometry (LDV) have gained increasing attention as a means to capture rich wavefield information directly from the structure and reduce dependence on uncertain a priori models [8, 28].

This paper validates a novel method based on empirical linear operators [5, 17] and a fundamentally periodic time formulation. The method yields a matrix-based description of the system and its inversion for defect localization [2]. We excite the structure with bonded piezoelectric transducers and use a LDV to perform a dense, one-time calibration scan that constructs source–receiver impulse-response operators. Wave propagation is represented on a discrete cyclic time axis so that it is modeled as circular convolution, diagonalized by the discrete Fourier transform (DFT). By designing the excitation to be strictly periodic and ensuring that no other sources act on the object, the cyclic time embedding is valid by construction. We impose periodic behavior in the experimental setup, making the frequency-domain treatment equivalent to the time-domain treatment. Within a single-scattering formulation, defect localization is solved via a Tikhonov-regularized inverse [29, 12] using only sparse monitoring data at transducer locations. We validate the approach experimentally on an anisotropic CFRP plate.

2. Mathematical framework

2.1. System assumptions

We assume: (i) linearity, (ii) time invariance, (iii) reciprocity, (iv) practical flatness (LDV access), and (v) the ability to control excitation relative to noise.

Assumptions (i), (ii), and (iii) are generally satisfied for mechanical waves under standard operating conditions. Assumption (iv) refers to geometries in which thickness is small relative to lateral dimensions, as is typically the case for plate-like components such as aircraft fuselage panels and shell structures. Assumption (v) requires controlled excitation, which is standard in active NDT/SHM configurations using piezoelectric or electromagnetic transducers.

2.2. Terminology and Notation

We use discrete-time vectors and matrices throughout. The spatial domain is discretized into a set of points indexed by integers, without implying any specific geometric arrangement. For example, a 100×100 scan grid of a square plate yields 10,000 points labeled 1 through 10,000, with an arbitrary but fixed mapping.

When a transducer emits a signal at location i, represented by the time-domain vector $\vec{s}_i$, the observed signal $\vec{m}_j$ at location j is given by convolution with the impulse response $\vec{g}_{ij}$ between those locations: $\vec{m}_j = \vec{g}_{ij} * \vec{s}_i$. This formulation separates spatial indices (subscripts) from the time-domain representation (vectors).

Under the reciprocity assumption, $\vec{g}_{ij} = \vec{g}_{ji}$. This relation applies to all variables in this paper with two spatial subscripts.

For computation, these vectors correspond to discrete-time samples over a fixed acquisition window, with all operations respecting the imposed periodicity described in subsequent sections.

2.3. Periodic time and discrete representation

Repeating the excitation enforces periodicity, allowing the time axis to be treated as a discrete cyclic domain. The impulse response between locations i and j can then be written as a circular-convolution operator $\boldsymbol{G}_{ij}$, represented by a circulant (circular Toeplitz) matrix fully determined by its first column, i.e., the sampled impulse-response vector $\vec{g}_{ij}$. Explicitly, for $\vec{g}_{ij} = (g_0, g_1, \dots, g_{n-1})^T$,

$$\boldsymbol{G}_{ij} = \begin{pmatrix} g_0 & g_{n-1} & \cdots & g_1 \\ g_1 & g_0 & \cdots & g_2 \\ \vdots & \vdots & \ddots & \vdots \\ g_{n-1} & g_{n-2} & \cdots & g_0 \end{pmatrix}, \tag{1}$$

so that $\boldsymbol{G}_{ij}\vec{v} = \vec{g}_{ij} * \vec{v}$ (circular convolution). Such operators are diagonalized by the discrete Fourier transform (DFT), enabling component-wise multiplication in the frequency domain.

2.4. Empirical impulse responses

For a transducer s and a measurement point i, the response is modeled as a circular convolution:

$$\vec{m}_i = G_{is}\vec{s}_s = \vec{g}_{is} * \vec{s}_s \tag{2}$$

where G_{is} is circulant with first column $\vec{g}_{is}$.

Consider a transducer at location s. We periodically emit a test pulse, and the LDV measures the response over one period across the object. The empirical impulse response $\vec{g}_{is}$ relates the measured vibrational *responses* at s and i through Eq. 2. Note, it does not correspond directly to the electrical drive signal applied to the actuator, hence the term "empirical impulse response."

Using commutativity of convolution, Eq. 2 can be rewritten as

$$\vec{m}_i = \vec{s}_s * \vec{g}_{is} = S_s\vec{g}_{is}. \tag{3}$$

We estimate $\vec{g}_{is}$ via Tikhonov-regularized least squares [15, 29] from multiple noisy pulse acquisitions (five in this paper) for each sensor–grid pair.

Let $(\vec{s}_s^p, \vec{m}_{is}^p)_{p=1}^P$ denote P repeated acquisitions. The Tikhonov-regularized least-squares problem is

$$\min_{\vec{g}_{is}} \sum_{p=1}^{P} \|\vec{m}_i^p - S_s^p\vec{g}_{is}\|_2^2 + \alpha\|\vec{g}_{is}\|_2^2. \tag{4}$$

The solution is computed efficiently, component-wise, in the frequency domain. For each frequency bin f (hats denote DFT components), Eq. 4 becomes

$$\min_{\hat{g}_{is}(f)} \sum_{p=1}^{P} \left|\hat{m}_i^p(f) - \hat{s}_s^p(f)\hat{g}_{is}(f)\right|^2 + \alpha|\hat{g}_{is}(f)|^2 \tag{5}$$

and is solved with regularization parameter $\alpha > 0$. The closed-form solution is

$$\hat{g}_{is}^{\text{reg}}(f) = \frac{\sum_p \overline{\hat{s}_s^p(f)}\,\hat{m}_i^p(f)}{\sum_p |\hat{s}_s^p(f)|^2 + \alpha}. \tag{6}$$

The resulting stabilized impulse response does not require an explicit noise model. We use a small α (typically $\approx 10^{-7}$) to avoid singularities, and the final solution is largely insensitive to the exact value.

2.5. Defect model

Since defects also satisfy the assumptions in Section 2.1, their effects can be modeled using circular convolutions. Within the single-scattering (Born-type) approximation, the scattering process is illustrated in Fig. 1. We assume a *delta-like* temporal reflection kernel, so each defect is represented by a scalar amplitude d_k at its spatial location. Algebraically, this corresponds to selecting only the first canonical temporal basis vector, $\vec{d}_k := d_k\vec{e}_0 = (d_k, 0, 0, \dots)^T$. Physically, it corresponds to a defect reacting instantaneously with amplitude d_k. See Section C.4 for a generalization of this approach.

2.6. Operator assembly

For an emitter at position j and a measurement at position i, the measured signal consists of a direct propagation term and first-order (single-scattered) contributions, as illustrated in Fig. 1:

$$\vec{m}_i = \vec{g}_{ij} * \vec{s}_j + \sum_{k=1}^{N} \vec{g}_{ik} * \vec{d}_k * \vec{g}_{kj} * \vec{s}_j. \tag{7}$$

The sum runs over all points on the plate, approximated by a finite set (e.g., a scan grid). Here, “single scattering” means first order in *defect-induced* scattering. Boundary reflections and other multipath propagation of the undamaged plate

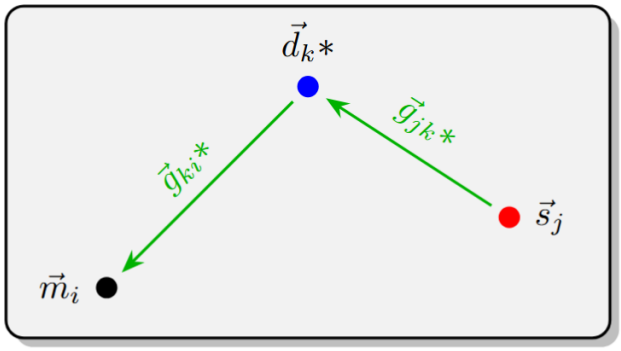


Figure 1: Schematic of a source at location j propagating to a defect at location k, which then re-emits the wave. The scattered wave subsequently propagates to a receiver at location i. In equation form, this process is $\vec{m}_i = \vec{g}_{kj} * \vec{d}_k * \vec{g}_{ik} * \vec{s}_j$.

are already embedded in the empirically measured impulse responses $\vec{g}_{ij}$; only terms with multiple defect interactions are neglected.

This can be rewritten in matrix form:

$$\vec{m}_i = G_{ij}\vec{s}_j + \sum_{k=1}^{N} G_{ik} D_k G_{kj} \vec{s}_j. \tag{8}$$

Using commutativity of circular convolution, we restate the problem as

$$\Delta\vec{m}_{ij} := \vec{m}_i - G_{ij}\vec{s}_j = \sum_{k=1}^{N} (G_{ik} G_{kj} S_j)\, \vec{d}_k. \tag{9}$$

Collecting the defect vectors $\vec{d}_k$ into a single vector gives

$$\vec{\mathbf{d}}_{\text{full}} := \begin{bmatrix} \vec{d}_1 \\ \vec{d}_2 \\ \vdots \\ \vec{d}_N \end{bmatrix}. \tag{10}$$

The problem can then be written in the general matrix form

$$L_{ij}^{\text{gen}} := \left[\, G_{i1}G_{1j}S_j \;\middle|\; G_{i2}G_{2j}S_j \;\middle|\; \cdots \;\middle|\; G_{iN}G_{Nj}S_j \right], \tag{11}$$

so that

$$\Delta\vec{m}_{ij} = L_{ij}^{\text{gen}}\, \vec{\mathbf{d}}_{\text{full}}. \tag{12}$$

This system is typically highly underdetermined and, even when it is not, often too large for practical computation. In our use case, the fully time-resolved defect vector would have dimension $N_{\text{grid}} N_t \approx 10^6$ (with N_{grid} spatial grid points and N_t time samples per period).

Assuming that scatterers generally re-emit waves almost immediately, we restrict $\vec{d}_k$ to a small set of temporal basis functions. As in the previous section, we use the simplest case, $\vec{d}_k = d_k \vec{e}_0$ (a one-dimensional basis). Since each defect location is parameterized by a single scalar d_k, the defect vector becomes

$$\vec{\mathbf{d}} := \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_N \end{bmatrix} \tag{13}$$

and is incorporated by including the basis vector in the operator:

$$L_{ij} := \left[\, G_{i1}G_{1j}S_j\vec{e}_0 \;\middle|\; G_{i2}G_{2j}S_j\vec{e}_0 \;\middle|\; \cdots \;\middle|\; G_{iN}G_{Nj}S_j\vec{e}_0 \right], \tag{14}$$

where $\vec{e}_0$ is understood as a column-vector-shaped matrix. The resulting problem statement for one emitter–sensor pair is

$$\Delta\vec{m}_{ij} = L_{ij}\vec{\mathbf{d}}. \tag{15}$$

For multiple emitter–sensor pairs, stacking the selected (i, j) pairs yields the combined system. For example, for one emitter at 1 and two sensors at 2 and 3:

$$\vec{\mathbf{m}}_{\text{full}} = \begin{bmatrix} \Delta\vec{m}_{12} \\ \Delta\vec{m}_{13} \end{bmatrix} \qquad L_{\text{full}} = \begin{bmatrix} L_{12} \\ L_{13} \end{bmatrix} \qquad \vec{\mathbf{m}}_{\text{full}} = L_{\text{full}}\vec{\mathbf{d}} \tag{16}$$

This corresponds to a $[[1, 2], [1, 3]]$ system. The general case can be much larger. In the examples presented here, five transducers yield ten emitter–sensor pairings (see Appendix C.2 for details).

We have therefore reduced the problem to the linear system in Eq. (16). The only approximations are spatial discretization, the first-order-scattering assumption, and the immediate-reflection approximation for defects.

2.7. Regularized inversion

We solve for $\vec{\mathbf{d}}$ in Eq. 16. Let $\mathcal{L} = L_{\text{full}}$ and $\mathbf{m} = \mathbf{m}_{\text{full}}$ from Eq. 16. The problem is the linear inversion $\vec{\mathbf{m}} = \mathcal{L}\vec{\mathbf{d}}$, which we solve by fidning the Tikhonov-regularized $\vec{\mathbf{d}}$ through minimizing

$$\operatorname{argmin}_{\vec{\mathbf{d}}} \; \|\mathcal{L}\vec{\mathbf{d}} - \vec{\mathbf{m}}\|_2^2 + \lambda\|\Gamma\vec{\mathbf{d}}\|_2^2 \tag{17}$$

where Γ specifies the penalty metric (a quadratic norm matrix) and $\lambda > 0$ controls the trade-off between data fit and stabilization. The normal equations yield the closed-form solution [15, 13]

$$\vec{\mathbf{d}} = (\mathcal{L}^\dagger\mathcal{L} + \lambda\Gamma)^{-1}\mathcal{L}^\dagger\vec{\mathbf{m}}. \tag{18}$$

The solution can be computed iteratively, or by precomputing and storing $(\mathcal{L}^\dagger\mathcal{L} + \lambda\Gamma)^{-1}\mathcal{L}^\dagger$ (more expensive up front). This enables rapid evaluation of $\vec{\mathbf{d}}$ for many new measurement sets $\vec{\mathbf{m}}$ via matrix–vector multiplication.

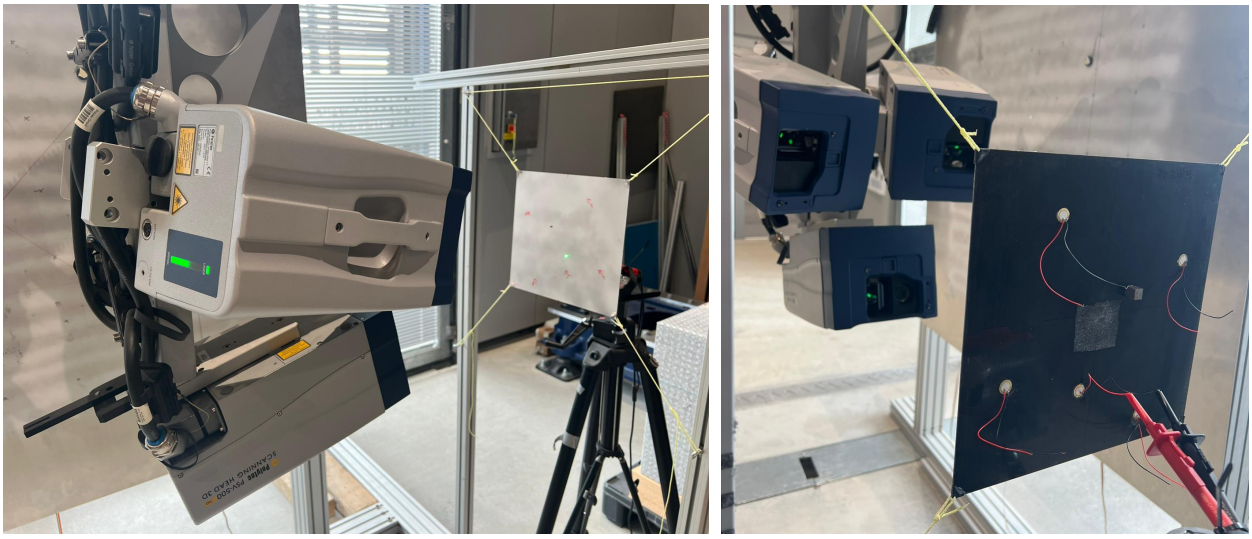
If Γ is the identity matrix, we recover standard Tikhonov regularization, which penalizes the Euclidean norm $\|\vec{\mathbf{d}}\|_2^2$. In practice, however, real systems can exhibit small changes in pulse shape and amplitude between calibration and monitoring. The inversion may then explain these differences by placing strong "defects" directly at or near transducer locations, which contaminates the rest of the reconstruction.

To isolate this effect, we choose Γ as a diagonal matrix with entries set to one everywhere except at grid points corresponding to transducers and in a small neighborhood around them, where entries are set to zero (unpenalized regions). These regions are therefore more likely to show up as defects, while remaining decoupled from the rest of the defect reconstruction. For visualization, these regions are masked out (set to zero).

Sophisticated strategies for selecting the optimal regularization parameter λ are discussed in [16], but they were not required for the present experimental and numerical examples. Here, λ is selected by a simple trial-and-error sweep over a small set of candidate values, typically $\lambda \in \{10, 100, 1000\}$. Because all measured amplitudes are normalized as described in Section 3, suitable values generally lie in this range.

3. Experimental Setup

The test specimen is a 26.8 cm × 26.8 cm × 0.14 cm bidirectional CFRP plate (8 layers, $[0°_2/90°_2]_s$, HexPly 8552/AS4). Five low-cost brass–ceramic piezoelectric discs (12 mm diameter) were bonded to the plate using Crystalbond adhesive. As shown in Section 2.4, the method is independent of the source transfer function and therefore accommodates a wide range of unknown coupling properties. The actuators were driven by a periodic pulse train with a 2 ms repetition period using an 80 kHz Ricker wavelet. A 3D LDV scanned the plate on a 104 × 104 square grid,



(a) (b)

Figure 2: Experimental setup for CFRP plate measurements. (a) Front view during LDV scanning; a white speckle coating is applied to increase surface reflectivity. (b) Rear view showing the five bonded transducers and the 1 cm^3 metal cube used as a defect proxy; the central square mark is residue from prior tests.

recording the full three-component velocity vector over time. In this work, we use only tFigure 3a shows one frame of the measurement. he out-of-plane V_z component, which is most strongly excited by the piezoelectric sources. The potential use of all three velocity components is discussed in Appendix A.1. The scan was repeated for each of the five transducers, constituting the one-time calibration step. Data were sampled at 625 kHz and low-pass filtered to 50 kHz prior to computation. Each measurement grid location was acquired with 10 averages in each of 5 separate runs, yielding five measurement sets. All velocities were normalized by dividing by the global maximum recorded velocity. This normalization improves numerical conditioning and helps keep suitable regularization parameters within a similar range across systems (Section 2.7). Consequently, plotted velocities (e.g., Figure 3a) are unitless and may not be directly comparable across figures (e.g., Figure 9b) when different scripts use different global maxima. The normalization does not affect the inversion result.

After calibration, a 1 cm^3 iron cube was bonded to the plate to simulate a defect. During monitoring, each transducer was excited again, but measurements were taken only at transducer locations. In our setup, the LDV acquired these sparse responses at the transducer positions.

To assess characteristic wavelengths relative to defect dimensions, the empirical dispersion relation of the CFRP plate was estimated from the calibration data (see Appendix A.2).

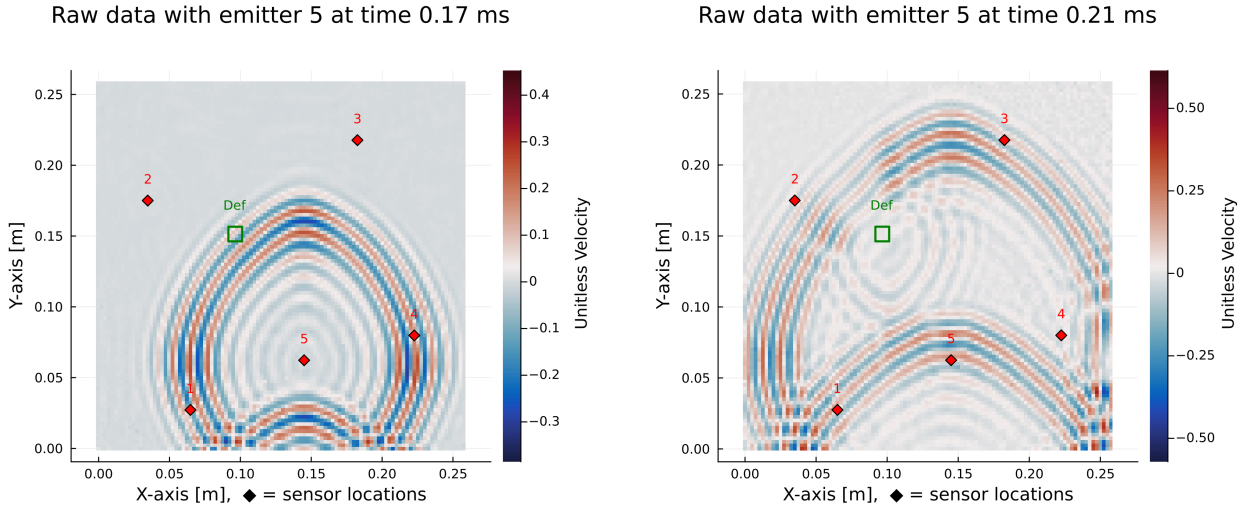
4. Results and Discussion

This section presents the validation and evaluation of the proposed empirical operator framework through a sequence of numerical and experimental case studies. Rather than separating observations from interpretation, each subsection examines a specific regime relevant to non-destructive testing (NDT) and structural health monitoring (SHM), highlighting both performance and practical limitations under controlled and realistic conditions.

Details of the sensor configurations are provided in Appendix C.2.

4.1. Baseline validation in a controlled numerical setting

We first assess the method in a controlled numerical environment to establish baseline performance under conditions consistent with the modeling assumptions. Elastic wave propagation is simulated using the spectral-element modeling suite Salvus [22] for a 50 cm × 50 cm aluminum plate with a thickness of 9 mm. A localized tapered inclusion with 1.5 cm diameter and relative density 0.2 is introduced to represent a small defect.



(a) (b)

Figure 3: Snapshots of the measured wavefield in the CFRP plate, low-pass filtered to 150 kHz. (a) Calibration scan at $t = 0.17$ ms without the defect present (location indicated by green square). (b) Wavefield at $t = 0.21$ ms after bonding the defect; this scan is shown for visualization only and is not used by the inversion. Red diamonds mark the five transducers (numbered 1–5), with transducer 5 active in this example. The reconstructions in this paper use 1–50 kHz bandpass filtering, which alters the pulse shape relative to the images shown here (see Appendix A.3).

The excitation consists of a periodic pulse train with a 2 ms repetition period using a 50 kHz Ricker wavelet. Five transducers are arranged in a configuration comparable to the experimental setup, and simulated measurements include 20% Gaussian RMS noise. The simulation is sampled on a 101×101 square grid over the plate. The data are low-pass filtered to 0–50 kHz to match the bandwidth used in subsequent experimental evaluations. The system matrix (Eq. 16) in this case has shape $10{,}201 \times 1{,}000$, meaning there are more than ten times as many defect parameters as measurement dimensions.

Figure 4a shows the reconstructed defect amplitude alongside the ground-truth density distribution. The defect is clearly localized at the correct position, demonstrating that the empirical operator formulation recovers small scatterers in a simulated environment despite the high degree of underdetermination. Background artifacts remain limited, indicating stable inversion behavior under controlled conditions.

This baseline result establishes a reference case against which deviations observed in experimental data and more challenging numerical scenarios can be interpreted.

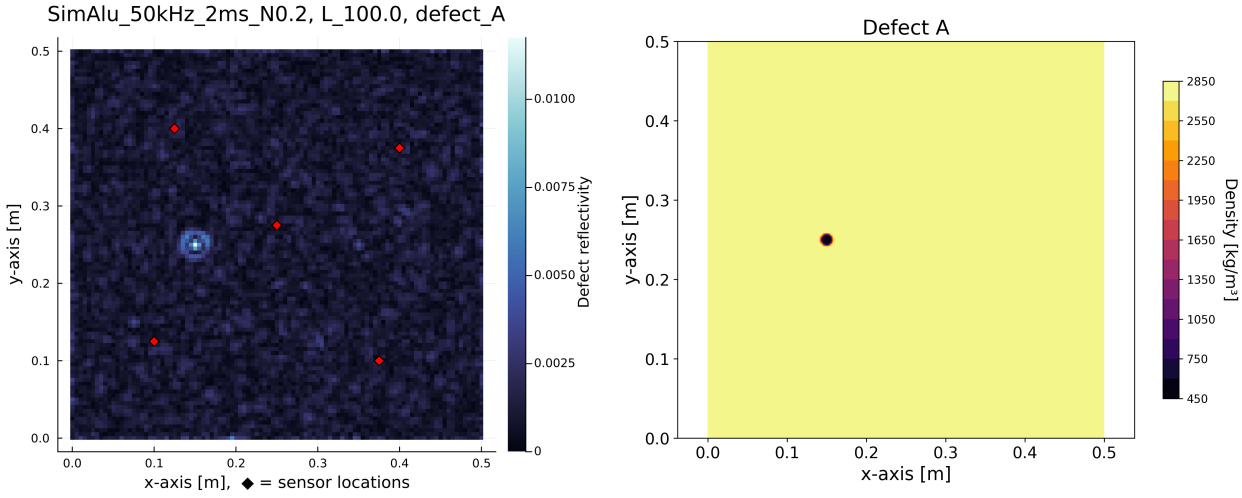
4.2. Experimental defect localization with sparse measurements

We next evaluate the method experimentally using the CFRP plate described in Section 3, thereby testing the calibration-to-monitoring workflow under real conditions. Following a one-time LDV calibration scan, defect localization is performed using only sparse measurements acquired at five transducer locations.

The experimental parameters closely follow the baseline simulation, with a 2 ms repetition period and excitation by an 80 kHz Ricker wavelet. Due to measurement noise at very low frequencies, the particle-velocity data are bandpass filtered to 1–50 kHz. An iron cube ($1 \times 1 \times 1$ cm^3) bonded to the plate serves as a defect proxy.

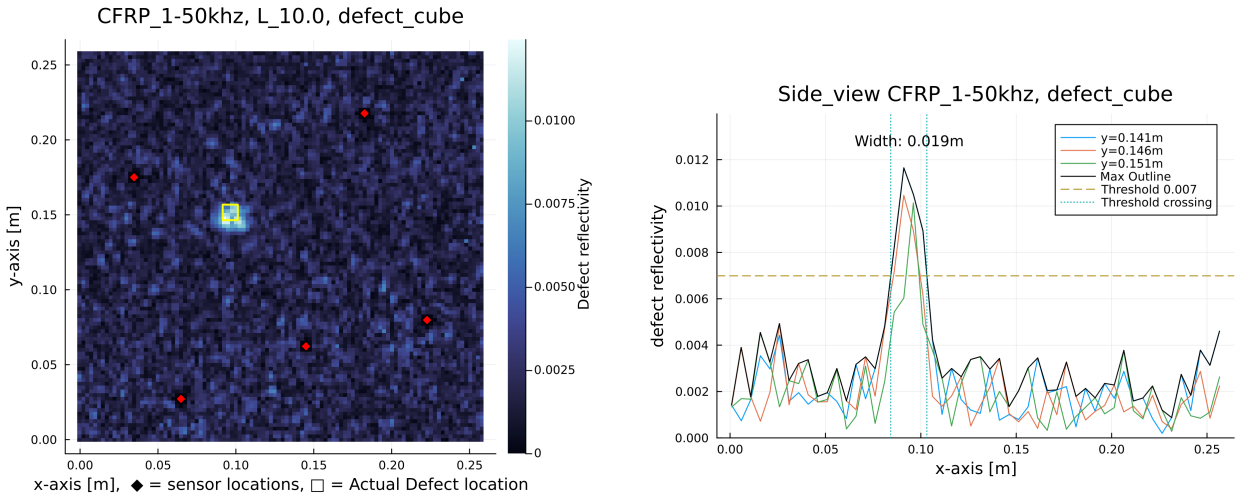
Figure 5a shows clear localization at the defect position (yellow square). Cross-sectional profiles through the reconstruction along the x-axis are presented in Figure 5b. Envelope slices exceeding 0.6 times the maximum amplitude yield an estimated localization half-width of approximately 2 cm. The minimum wavelength in the 1–50 kHz band is approximately 1.25 cm (see Appendix A.2 for the derivation).

Small black regions surrounding the transducers (red diamonds) are visible in Figure 5a. These correspond to exclusion zones defined by the unpenalized region in the regularization (Section 2.7) and are masked in the visualization. Without masking, these regions exhibit elevated amplitudes primarily caused by systematic experimental



(a) (b)

Figure 4: Simulation case A with a single small inclusion. (a) Reconstructed defect amplitude in the 0–50 kHz band obtained using the empirical-operator inversion, demonstrating accurate localization of a 1.5 cm inclusion under 20% RMS noise. (b) Ground-truth density map showing the tapered circular lower-density inclusion (relative density 0.2).

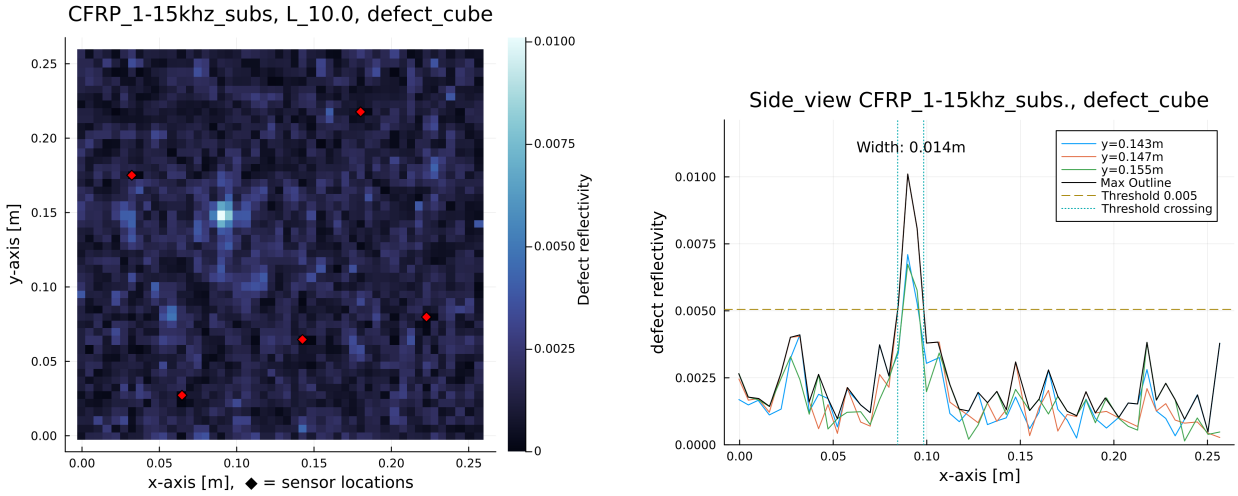


(a) (b)

Figure 5: Defect localization results in the 1–50 kHz band using sparse transducer measurements. (a) Reconstructed magnitude of the defect vector $|\vec{\mathbf{d}}|$ for the experimental CFRP plate, showing clear localization of the adhered metal cube near its true position (yellow square). Red diamonds indicate the five sensor locations; black regions surrounding them correspond to unpenalized exclusion zones introduced by the regularization. (b) Cross-sectional profiles along the x-axis through the reconstruction. Colored envelopes denote slices exceeding 0.6 times the maximum amplitude, yielding an estimated localization half-width of approximately 2 cm.

differences, such as pulse-amplitude variations between calibration and monitoring, rather than by physical defect scattering.

The experimental result demonstrates that, once calibrated, the framework enables reliable defect localization using sparse monitoring measurements only. This separation between a data-intensive calibration stage and a lightweight monitoring stage is central to the intended SHM workflow and motivates further investigation of the method's behavior across different frequency regimes and modeling assumptions.



(a) (b)

Figure 6: Defect localization results in the 1–15 kHz band using sparse transducer measurements and a 50×50 subsampled reconstruction grid. (a) Reconstructed magnitude of the defect vector $|\vec{\mathbf{d}}|$ for the experimental CFRP plate. Red diamonds indicate the five sensor locations; the adhered metal-cube defect is clearly resolved (marker omitted for visual clarity; see Figure 5a). (b) Cross-sectional profiles along the x-axis through the reconstruction. Colored envelopes denote slices exceeding 0.5 times the maximum amplitude, yielding an estimated localization half-width of approximately 1.4 cm.

4.3. Localization at very low frequencies (experimental limit)

To investigate the low-frequency limit of the method, we evaluate the experimental dataset in the 1–15 kHz band, corresponding to the audible frequency range. In this regime, wavelengths approach the characteristic dimensions of the plate, providing a stringent test of localization capability under reduced information content. The matrix shape here (2503, 290), resulting is a similar under determination ratio as both previous results.

Applying the inversion on the full 104×104 reconstruction grid yields mixed results (Appendix A.4), characterized by unstable reconstructions and isolated high-amplitude pixels. To improve conditioning, the inversion is therefore restricted to a 50×50 spatial subgrid while using the same measurement data. The resulting reconstruction, shown in Figure 6, exhibits a compact defect signature with an estimated width at half maximum of approximately 1.4 cm. At a minimum wavelength of roughly 1.7cm, the localization is slightly below wavelength size.

4.4. Multiple-defect retrieval and limits of single scattering

We next examine a scenario that departs from the single-defect baseline by introducing two tapered inclusions with diameters of 1.5 cm and 5 cm at relative density 0.2 (Figure 7). All simulation parameters are identical to those used in the baseline configuration.

The reconstruction successfully localizes both defects, demonstrating that the method can distinguish multiple scatterers and recover larger structures despite internally relying on a single-scattering formulation. However, pronounced background artifacts are observed compared to the single-defect case. While this trend is consistent with increased scattering complexity, the present simulation does not isolate higher-order defect interactions as the sole cause. The larger and stronger scatterer likely contributes substantially to the observed background structure.

This case therefore illustrates a practical limitation of the current formulation: while localization remains feasible, reconstruction quality decreases for more challenging multi-defect configurations. In continuous monitoring scenarios where defects emerge successively, this limitation may be partially mitigated by periodic recalibration (or matrix updates) that incorporate the current structural state. Under such incremental updates, a local single-scattering approximation can remain appropriate for each monitoring interval.

4.5. Resolution dependence on temporal bandwidth

To investigate whether additional temporal information mitigates the degradation observed above, the same two-defect configuration is simulated with an increased repetition period of 5 ms and an extended frequency range of 0–100 kHz. Relative to the baseline configuration, the algorithm therefore has access to approximately five times more temporal information ($2\times$ higher frequency bandwidth and $2.5\times$ longer period), resulting in the matrix size being ($10'201 \times 2500$).

The reconstruction shown in Figure 8a exhibits improved fidelity and substantially reduced background artifacts compared to the previous case. The longer listening window allows additional scattered energy to be captured within a single period, while the increased bandwidth enhances temporal resolution, together improving the separability of scattering contributions from multiple defects.

These observations indicate that part of the degradation seen in the multi-defect scenario arises from limited temporal information rather than solely from violations of the single-scattering assumption. Increasing bandwidth and observation time therefore sharpens localization and suppresses artifacts, representing a near-idealized scenario in which information limitations are reduced. A systematic analysis of these dependencies is deferred to [33].

4.6. Single-transducer (pulse–echo) feasibility

Finally, we investigate a minimal sensing configuration in which a single transducer simultaneously acts as emitter and receiver, corresponding to a pulse–echo measurement geometry. This configuration represents an extreme reduction in spatial sampling and tests whether localization remains possible with strongly limited measurement aperture. This problem setting is related to those studied in [10, 9], enabling comparison with alternative approaches.

The simulation setup follows the baseline validation case section 4.1, using a 5 ms repetition period and a 0–100 kHz bandpass filter with 10% RMS noise, while all other parameters remain unchanged. The resulting reconstruction is shown in Figure 8b.

Despite the severe reduction in measurement information (Matrix size $10'201 \times 500$), the defect remains clearly localized. The reconstructed signature more closely resembles the high-bandwidth multi-defect result in Figure 8a than the baseline low-bandwidth case (Figure 4a), suggesting that localization performance is primarily governed by available frequency content rather than sensor count alone. However, more complex defect configurations reconstruct poorly in this pulse–echo geometry, and the solution becomes more sensitive to noise and degrades substantially at lower frequencies.

These observations indicate that while sparse or even single-sensor implementations are feasible in principle, their robustness depends strongly on bandwidth and signal quality. The behavior is consistent with findings reported in [17], where increased sensor counts mainly reduce background artifacts rather than fundamentally improving spatial resolution. Attempts to apply this configuration to real-world measurements did not yield interpretable results, consistent with the simulated sensitivity under comparable low-frequency and short-period constraints.

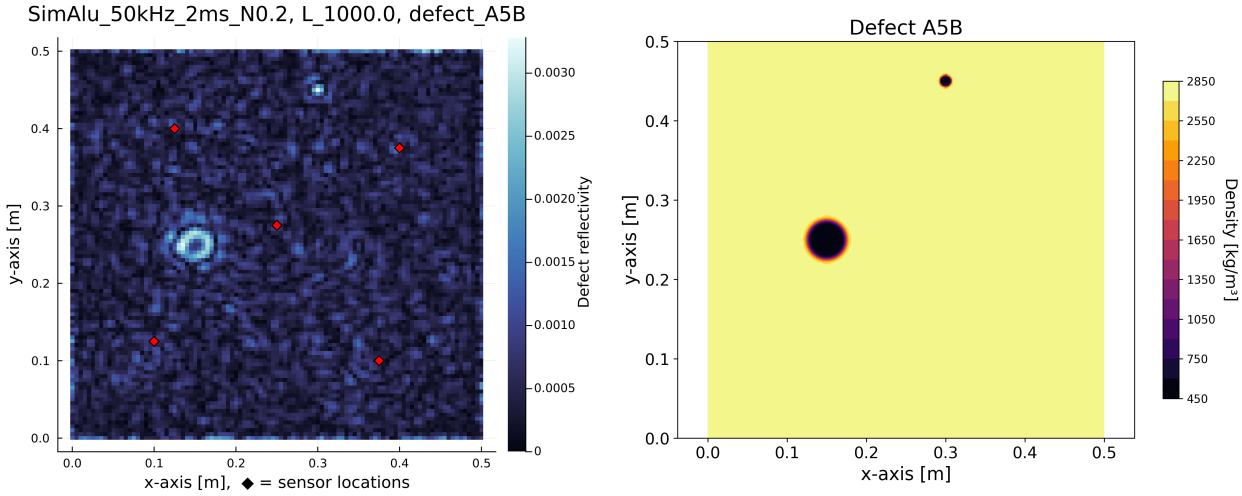
4.7. Practical implications and limitations for SHM

Across the numerical and experimental cases presented above, the empirical operator framework demonstrates consistent defect localization under sparse sensing conditions, while also revealing practical limits relevant for structural health monitoring (SHM) applications. The calibration-to-monitoring workflow enables a separation between a data-intensive initialization stage and rapid subsequent evaluations, suggesting suitability for repeated inspections once the system has been calibrated.

The reconstructed amplitudes throughout this work should be interpreted as relative indicators of defect location rather than quantitative scattering strengths, since the regularization parameter λ influences amplitude scaling. Future implementations could address this dependence by explicitly characterizing the amplitude bias introduced by regularization and by accounting for grid-density effects in the inversion.

A comparison between simulations and experiments indicates broadly consistent behavior under comparable bandwidth and sparsity conditions. The experimental configuration differs slightly by restricting the usable band to 1–50 kHz due to low-frequency measurement noise and by employing an adhered metal cube as a defect proxy rather than material density changes. Nevertheless, the agreement with the baseline simulation supports the numerical setup as representative of the method's behavior under controlled conditions.

Several limitations emerge from the combined case studies. Reconstruction quality degrades when scattering complexity increases or when the single-scattering approximation becomes less valid, as observed in the multi-defect simulations. Likewise, when the number of free spatial parameters vastly exceeds the effective information content of



(a) (b)

Figure 7: Simulation case A5B with two inclusions. (a) Reconstructed defect amplitude obtained using the empirical-operator inversion, showing simultaneous localization of the 1.5 cm and 5 cm inclusions together with increased background artifacts. (b) Ground-truth density map containing two tapered circular lower-density inclusions (relative density 0.2).

the data, reconstructions may become unstable, particularly at low frequencies. Accurate recovery of defect shape is not expected when defect dimensions are significantly smaller than the wavelength [33].

The empirical operators are calibrated on the baseline structure and therefore encode direct paths together with stable boundary/multipath reflections. In weakly bounded or near-unbounded media, this reduced propagation diversity can make sparse-sensor localization less robust. Conversely, complex but stable scattering backgrounds (e.g., boundaries or pre-existing features) can improve localization of newly emerging defects. If the background changes substantially, recalibration or operator updates are required.

Finally, the experimental defect differs from realistic CFRP damage such as delaminations. However, it serves as a controlled proxy for contact-based perturbations. Together with the low computational cost after calibration (Appendix C.1), the results suggest that the framework may enable practical SHM implementations based on sparse sensing, and potentially low-cost wave-based interfaces such as those explored in [17].

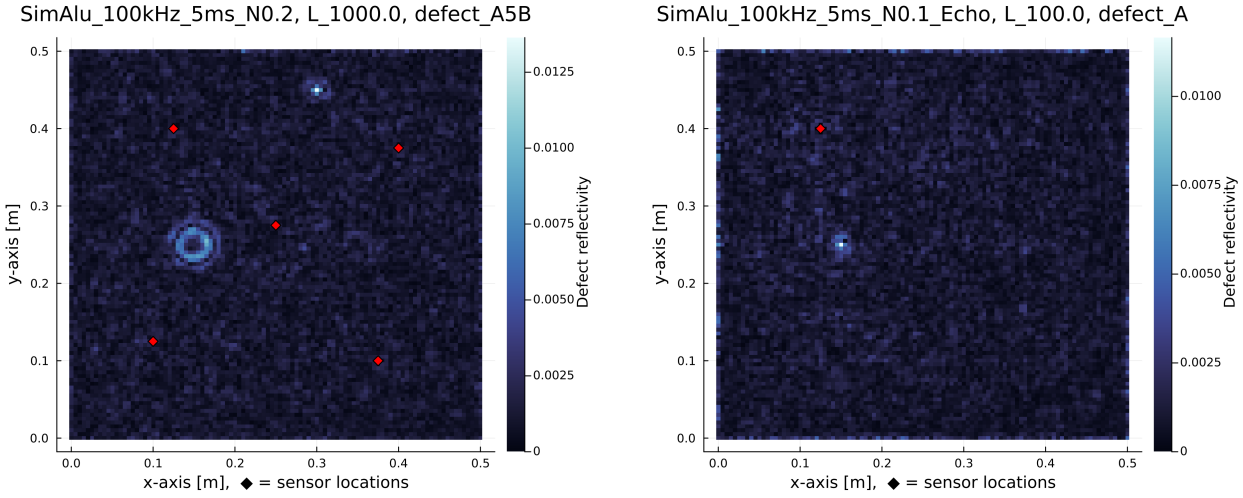
5. Conclusion

This work presented and experimentally validated an empirical, fully vectorized linear-operator framework for inverse scattering motivated by structural health monitoring applications. Source–receiver impulse-response operators are constructed using bonded low-cost piezoelectric transducers and a one-time LDV calibration scan. By enforcing a discrete periodic time axis, wave propagation is represented as DFT-diagonalizable circular convolution, allowing defect localization to be formulated as a Tikhonov-regularized linear inverse problem.

Experimental validation on an anisotropic CFRP plate demonstrated defect localization from sparse measurements acquired only at transducer locations after calibration. An adhered 1 cm^3 iron cube was localized yielding a characteristic width of about 2 cm at a wavelength of approximately 1.2 cm. Localization remained observable even in the very-low-frequency 1–15 kHz band. Complementary simulations reproduced these observations under controlled conditions and enabled evaluation of practical limits, including multiple-defect scenarios, bandwidth dependence, and pulse–echo operation.

Performance degrades when the single-scattering assumption becomes less valid or when the number of reconstruction parameters vastly exceeds the information supported by the data. Additional practical limitations arise from the simplified delta-like defect representation and the reliance on LDV measurements during calibration, which may introduce sensitivity to experimental alignment and measurement conditions.

The defect reconstruction fidelity appear to be limited by the maximum frequency available, while sensor counts, and more total information available increases stability.



(a) (b)

Figure 8: Simulation results illustrating the influence of increased temporal bandwidth and reduced sensor count. (a) Reconstruction of case A5B using a 5 ms repetition period and 0–100 kHz bandwidth, showing reduced background artifacts and sharper localization compared to Figure 7. (b) Single-transducer pulse–echo reconstruction of the small defect from case A (Figure 4), demonstrating successful localization using a 0–100 kHz bandwidth with 10% RMS noise.

The results demonstrate that reliable defect localization can be achieved using sparse measurements once empirical operators have been established. Requiring only linear time-invariant system behavior and no explicit forward model, the framework offers a data-driven alternative for applications where maintaining accurate digital twins is impractical. The demonstrated calibration-to-monitoring workflow provides a pathway toward practical SHM implementations enabling rapid, low-cost monitoring and potentially extending to other wave-based sensing and interaction systems.

CRediT authorship contribution statement

R. G. Wueest: Conceptualization, Software and Evaluation, Writing – original draft. **J. Aichele:** Supervision, Methodology, Investigation, Writing - review & editing. **J. Mueller:** Supervision, Discussion, Writing - review & editing. **H. R. Thomsen:** Supervision, Methodology, Investigation, Resources, Writing - review & editing.

Declaration of generative AI in scientific writing

During the preparation of this manuscript, the authors used a generative artificial intelligence tool to assist with language refinement, structural editing, and clarity improvements. The authors reviewed and edited the output as necessary and take full responsibility for the content of the publication.

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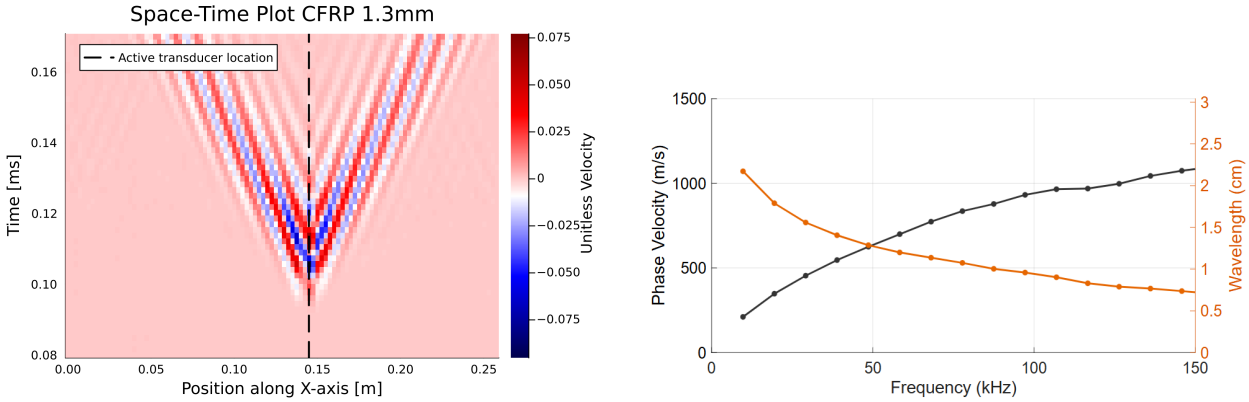


(a) (b)

Figure 9: Empirical dispersion of the CFRP plate obtained from the calibration scan. (a) Space–time diagram along the $y = 6.2$, cm line through the active transducer (no. 5), derived from the dataset shown in Figure 3a. (b) Phase velocity v_p (black, left axis, m/s) and wavelength λ (orange, right axis, cm) as functions of frequency f (kHz); values are approximate due to material anisotropy.

A. Additional experimental details and preprocessing

This appendix summarizes experimental processing choices relevant for interpreting the results, including the selected measurement component, filtering effects introduced by the periodic-time formulation, and auxiliary low-frequency evaluations.

A.1. Longitudinal velocity components

Only the out-of-plane LDV velocity component V_z is used throughout this work. This component is excited most strongly by the bonded piezoelectric transducers and provides the highest signal-to-noise ratio in the experimental dataset.

The proposed framework itself is not restricted to a specific wave mode or measurement component, as it operates on an empirically calibrated linear time-invariant input–output relation. In principle, any measured observable (e.g., V_x, V_y, V_z, or transducer voltage) could be used. However, in the present experiments the in-plane components (V_x, V_y) exhibited substantially lower SNR and contained numerous unreliable or missing grid points, resulting in non-interpretable reconstructions. For this reason, all experimental results are restricted to V_z.

A.2. Empirical dispersion analysis

To obtain approximate wavelength scales for interpreting the localization results, the dispersion properties of the CFRP plate were estimated from the calibration dataset. An x–t slice (Figure 9a) was extracted along the x-axis at $y = 6.2$ cm through the active transducer from the dataset shown in Figure 3a.

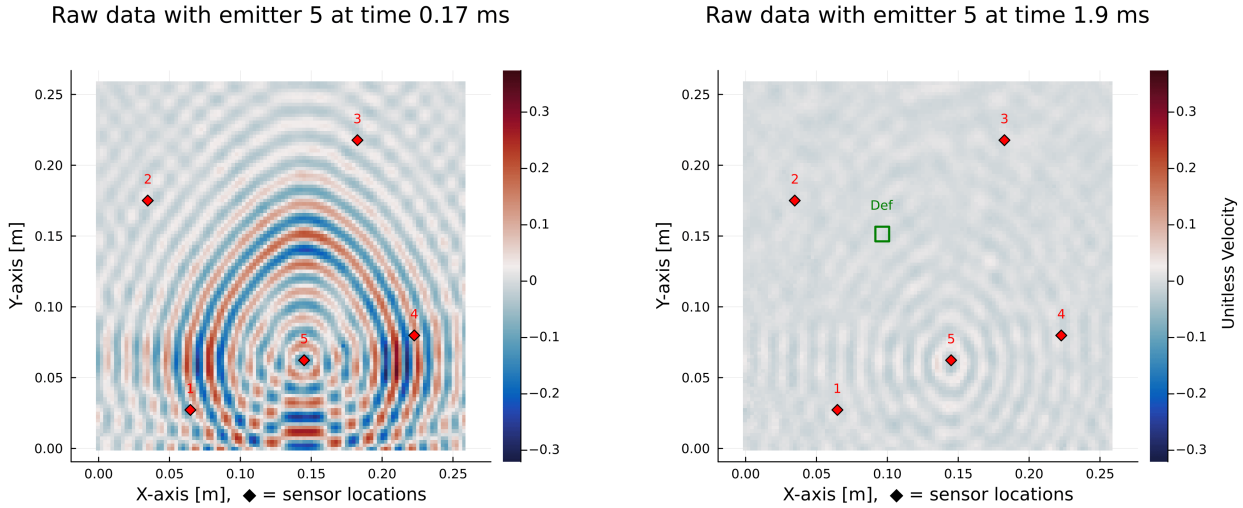
Because the spatial sampling of the measurement grid is not perfectly uniform, estimating dispersion curves using a 2D FFT is unreliable. Instead, the RANSAC-based method described in [1] was applied to robustly fit propagation slopes in the x–t diagram and estimate the corresponding phase velocities. The resulting phase velocities and wavelengths are shown in Figure 9b.

These values should be regarded as approximate. The analysis assumes a uniform spatial grid, which slightly deviates from the actual sampling. In addition, the true wavelengths vary with propagation direction due to material anisotropy.

A.3. Effect of 1–50 kHz filtering under periodic time

The results in the main text employ a 1–50 kHz bandpass filter, whereas the calibration snapshot shown in Figure 3a is effectively unfiltered at 150 kHz. Figure 10(a) shows the same acquisition time after filtering. The reduced bandwidth increases the effective pulse duration, causing the response to extend over a larger fraction of the periodic time window.

As a consequence, near the end of the 2 ms period ($t \approx 1.9$ ms), the tail of the current pulse overlaps with the onset of the next repetition, as illustrated in Figure 10(b). This apparent wraparound is a direct consequence of the discrete



(a) (b)

Figure 10: Effect of 1–50 kHz bandpass filtering on the transducer pulse shape within the periodic time framework. (a) CFRP plate response at $t = 0.17$ ms with transducer 5 active. Compared to the unfiltered snapshot in Figure 3a, the pulse appears broadened due to low-pass filtering. (b) Response at $t = 1.9$ ms, near the end of the 2 ms period. The tail of the current repetition and the onset of the next repetition are simultaneously visible, illustrating the wraparound inherent to the periodic time formulation.

periodic-time formulation used throughout, and is consistent with the periodic implementation of the experiments. The framework therefore remains applicable even when pulses are not temporally isolated within a single period.

A.4. Very-low-frequency evaluation (1–15 kHz and 1–30 kHz)

To further probe the low-frequency regime, the 1–15 kHz and 1–30 kHz bands were evaluated using the original full-resolution 104×104 defect grid (10 816 unknowns). From Figure 9, the corresponding minimum wavelengths are approximately 16 cm and 10 cm, respectively.

The reconstructions shown in Figure 11 demonstrate increasing instability at these frequencies. Although the defect region remains broadly identifiable, the images exhibit pronounced background artifacts and isolated high-amplitude pixels, particularly on the right side of the plate. In contrast, restricting the inversion to a reduced 50×50 grid in the main text (Figure 6) produces a stable and compact localization.

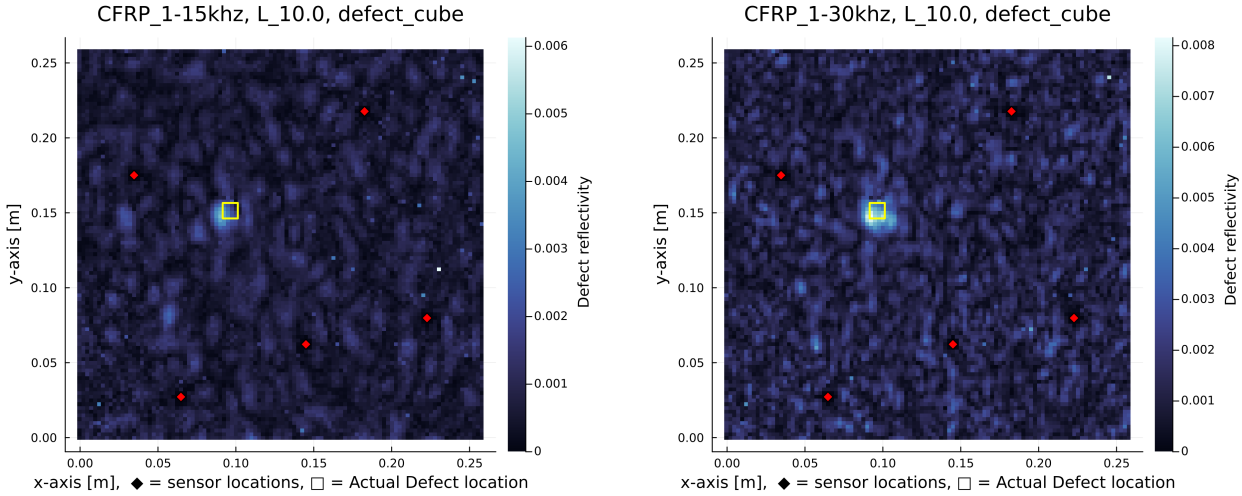
B. Additional diagnostics and limitations

This appendix summarizes secondary observations that influence reconstruction quality but are not central to the main methodological development.

B.1. High-frequency degradation in the experiment

In the experimental example of Section 4.2, extending the usable bandwidth beyond approximately 50 kHz provides only limited improvement, and reconstructions degrade for frequencies above 100 kHz. This behavior contrasts with the numerical simulations, where increased bandwidth consistently improves reconstruction quality.

The discrepancy is attributed to increased sensitivity of higher-frequency components to experimental imperfections, including calibration–monitoring misalignment, phase errors, and attenuation effects that are not fully captured by the empirical operator calibration. As a result, higher-frequency data contribute less useful information for defect localization in the present experimental configuration, whereas lower-frequency bands remain comparatively robust.



(a) (b)

Figure 11: Low-frequency defect reconstructions on the full 104×104 grid. (a) Reconstruction using a 1–15 kHz bandpass filter, showing isolated high-amplitude pixels and pronounced background artifacts. (b) Reconstruction using a 1–30 kHz bandpass filter on the same grid. Compared to the subsampled 50×50 result in Figure 6, the full-grid inversions exhibit pixel-scale spikes.

C. Implementation aspects and extensions

This appendix summarizes computational characteristics of the implementation and outlines possible extensions of the empirical operator framework beyond the configurations considered in the main text.

C.1. Computational performance

In the implementation used in this work, the global system matrix $\mathcal{L}_{\text{full}}$ from Eq. (14) is formed explicitly and inverted in a Tikhonov sense via

$$(\mathcal{L}^T \mathcal{L} + \lambda \Gamma)^{-1} \mathcal{L}^T.$$

For a matrix of size $n \times m$, where n denotes the number of measurements and m the number of defect parameters, memory requirements scale as $\mathcal{O}(nm)$ and inversion cost as $\mathcal{O}(nm^2)$. Increasing the number of measurements is therefore comparatively inexpensive, whereas enlarging the defect space dominates computational cost.

For the experimental CFRP case in Section 4.2 ($\sim 10^3$ measurements and $\sim 10^4$ defect parameters), the one-time inversion requires approximately 17 s on a standard laptop (Intel i7 CPU, 16 GB RAM). Computation of empirical impulse responses requires about 4 s and scales approximately linearly with grid size. Once calibration is completed, subsequent evaluations reduce to a single matrix–vector multiplication and require only 3–5 ms per update.

This separation between a moderately expensive calibration stage and rapid evaluation enables efficient repeated monitoring once the empirical operators have been established.

C.2. Sensor configurations

The sensor configuration used for all 5-sensor results in Section 4 is [[1,2], [1,3], [1,4], [1,5], [2,3], [2,4], [2,5], [3,4], [3,5], [4,5]]. For the echo configuration in Figure 8b, it is [[1,1]].

In hindsight, a layout such as [[1,2], [1,3], [2,3], [2,4], [3,4], ...] may better balance the roles of emitters. This was considered too late for the present study. Follow-up tests did not reveal noticeable improvements, and the original configuration was retained.

Measurement pairs were selected to avoid redundant reciprocal combinations (e.g., $[i, j]$ and $[j, i]$), which provide limited additional information under the reciprocity assumption. The presented configuration was found sufficient to obtain stable reconstructions while maintaining sparse sensing conditions.

C.3. Toward fully transducer-based monitoring

The present study relies on LDV measurements during calibration to obtain dense wavefield sampling. In principle, a fully transducer-based implementation could replace the LDV by combining emitting and sensing transducers during calibration, allowing empirical operators to be constructed directly from embedded sensors. Such an approach would extend the method toward practical structural health monitoring scenarios in which only permanently installed transducers are available.

C.4. Generalization to arbitrary defect bases

The results in this paper employ a delta-like defect basis. The formulation can be generalized by introducing a set of empirical basis vectors describing different defect responses. Let $\mathbf{b}_1, \mathbf{b}_2$ denote two such basis vectors and define the projector matrix

$$P := [\mathbf{b}_1 \ \mathbf{b}_2].$$

Replacing $\vec{e}_0$ in Eq. (12) yields the modified operator

$$P := \begin{bmatrix} \vec{b}_1 & \vec{b}_2 \end{bmatrix}. \tag{19}$$

Replacing $\vec{e}_0$ in Eq. 14, we obtain

$$L_{ij} := \left[\, G_{i1}G_{1j}S_jP \;\middle|\; G_{i2}G_{2j}S_jP \;\middle|\; \cdots \;\middle|\; G_{iN}G_{Nj}S_jP \right] \tag{20}$$

and the expanded defect vector

$$\mathbf{d} = \begin{bmatrix} d_{1,1} \\ d_{1,2} \\ d_{2,1} \\ \vdots \\ d_{N,2} \end{bmatrix},$$

where k indexes spatial location and the second index denotes the selected basis element.

This extension is not explored further here but may become important for representing distributed defects such as delaminations that deviate from delta-like scattering behavior.